

Reproducing and Extending KELE: A Multi-Agent Framework for Structured Socratic Teaching with LLMs

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Abstract

We reproduce and extend KELE (Peng et al., 2025), a multi-agent Socratic teaching framework. Our four contributions are: (i) a *fusion* architecture collapsing consultant and teacher onto one open-weight backbone via a JSON-schema-constrained structured-output call; (ii) a supervised classifier replacing the LLM consultant ($\sim 800\text{M}$ -parameter Qwen3.5-0.8B-LoRA, 67.57% test state accuracy) paired with a Gemma 4 31B teacher and 10-shot stage-balanced exemplars — our locked $n=681$ headline at 55.39% state accuracy / unified 72.24, a $2.14\times$ lift over the GPT-4o + SocratTeachLLM baseline; (iii) a **benchmark critique showing surface-form metrics systematically reward training-data memorization over teaching capability** (rankings invert; gap widens monotonically with n -gram length; English translation reproduces the published headline within 1.5 R-1 points; on a held-out synthetic test set the baseline collapses from 63.40 to 32.86 stage-balanced macro); and (iv) under a memorization-resistant unified score, our open-weight system overtakes the best frontier teacher we measured by +2.18 unified points at canonical sample size, on a single 32 GB consumer GPU at zero per-run eval API cost. We also release SocratDataset-EN and SocratDataset-SYNTHETIC. Code, datasets, and reproduction scripts are public.

1 Introduction

Problem and relevance. Socratic teaching is a pedagogy in which the teacher elicits answers through structured questioning rather than direct instruction (Seeskin, 1987; Chang et al., 1998). Research consistently shows it produces deeper learning outcomes than passive knowledge-imparting instruction (Johnston, 1994; Knezic et al., 2010), but it requires highly skilled instructors and does not scale — a one-to-many bottleneck that the educational use of large language models (LLMs)

could in principle relieve (Kasneci et al., 2023; Gan et al., 2023; Sonkar et al., 2023). Yet existing LLM tutoring systems treat Socratic dialogue mostly as turn-level heuristic questioning (Al-Hossami et al., 2024; Shridhar et al., 2022; Zhang et al., 2024; Dan et al., 2023; Liu et al., 2024; Ding et al., 2024; Macina et al., 2023); very few enforce the coherent multi-stage pedagogical progression a skilled human Socratic tutor follows. This is precisely the gap KELE (Peng et al., 2025) targets, and that we examine in this work.

The KELE framework. KELE decomposes Socratic teaching into five strictly-ordered stages (Student Questioning, Concept Probing, Inductive Reasoning, Rule Construction, Teacher Summary) governed by a rule system called SocRule containing 34 teaching strategies. A *consultant* agent classifies the student’s cognitive state and selects a strategy; a *teacher* agent then generates the dialogue response that implements it. KELE fine-tunes GLM4-9B (Team GLM et al., 2024) via LoRA (Hu et al., 2022) on the SocratDataset training split to produce **SocratTeachLLM**, a teacher model the authors report outperforms GPT-4o (OpenAI, 2024) across all nine Socratic-teaching evaluation dimensions despite being much smaller. The architectural insight — separation of teaching *planning* from teaching *execution* — mirrors real-world co-teaching practice (Nevin et al., 2009), where single-agent approaches conflate the two roles (Ge et al., 2023; An et al., 2024).

Motivation and goals. The published KELE result rests on surface-form text-overlap metrics (ROUGE (Lin, 2004), BLEU (Papineni et al., 2002)) — metrics designed for translation/summarization where the valid-output space is small. Socratic teaching has a much larger space of pedagogically equivalent responses (paraphrases of the same teaching move), so surface-form metrics conflate phrasing memorization with teaching qual-

Stage	Phase	States	Turns
a	Student Questioning	2	6,803
b	Concept Probing	6	7,480
c	Inductive Reasoning	22	14,445
d	Rule Construction	4	7,361
e	Teacher Summary	1	6,803
Total		35	42,892

Table 1: SocRule overview. Stage c dominates with 22 states and 33.7% of all turns.

ity. We set out to (1) reproduce KELE end-to-end, (2) extend it on a single-GPU consumer budget that forecloses the canonical two-model stack, and (3) test whether the published surface-form rankings hold up against memorization-resistant evaluation. Our open-weight integration architecture (§2) lifts overall state accuracy by **+29.45** percentage points (a **2.14** $\times$ multiplier) over the GPT-4o + SocratTeachLLM baseline, at 0 per-run API cost; the benchmark critique (§3) provides four converging lines of evidence that the published 9B specialist’s surface-form advantage is language-bound memorization rather than transferable pedagogical capability.

2 Proposed Method and Implementation

2.1 Datasets and the SocRule label space

SocratDataset (Peng et al., 2025) comprises 6,803 multi-turn Chinese dialogues totalling 42,892 annotated turns from elementary-school science education, with every turn labeled into one of 35 pedagogical states across the five SocRule stages (Table 1). Stage c (Inductive Reasoning) is structurally the hardest: it has 22 fine-grained states and contains 33.7% of all turns. We use the published 90/10 split (6,122 train / 681 test) throughout. As a secondary contribution we release two derivative datasets: **SocratDataset-EN**, a complete English translation of all 6,803 dialogues generated by a local Qwen3.5-9B instance with JSON-structure validation; and **SocratDataset-SYNTHETIC**, 75 Chinese dialogues generated end-to-end by Claude Sonnet 4.6 and verifiably disjoint from SocratDataset’s published splits but pedagogically matched, used as a clean-probe contamination control (§3).

2.2 Single-GPU budget forces an architectural pivot

All open-weight runs use a single NVIDIA RTX 5090 (32 GB VRAM). The canonical KELE

stack co-locating SocratTeachLLM (≈ 19 GB) with a separately served frontier consultant does not fit. We respond with two design choices: (i) a *fusion* variant in which both consultant and teacher are emitted by a single JSON-schema-constrained LLM call with fields {evaluation, state_code, teacher_response}, enforced server-side by llama.cpp’s strict json_schema grammar; and (ii) the deterministic-classifier integration described below, which replaces the consultant LLM with a supervised classifier and *removes the JSON-schema dependency from the consultant path entirely*. The schema dependency matters: at full $n=681$ scale, Qwen 35B-A3B’s fusion fallback rate is 0.91% (38/4,171 turns), but Gemma 4 31B’s standalone fusion fallback rate is 21.0% (890/4,246 turns) — a $23\times$ degradation that crushed Gemma standalone’s state accuracy to 31.39% (vs. a smoke-mini-projected 46.71%) and motivated retracting Gemma as a standalone candidate. The integration architecturally bypasses the issue. Reproduction of the canonical KELE stack uses GPT-4o (OpenAI, 2024) as consultant and SocratTeachLLM served via vLLM (Kwon et al., 2023).

2.3 Integration architecture

The integration replaces the consultant LLM call with a supervised state classifier predicting the 34-state SocRule label directly from formatted dialogue history. We evaluate two classifier architectures on SocratDataset’s 42K labeled turns.

(1) 24M-parameter Chinese BERT (BAAI/bge-small-zh-v1.5, ~ 95 MB resident): the 34-state head fine-tunes in 148 s and reaches **61.64%** test state accuracy. A complementary 5-way stage-classification head from the same backbone trains in only 92 s to **86.55%** stage accuracy — **+75** percentage points over GPT-4o on stage c (the 22-way inductive-reasoning classification), and the strongest result on the hardest single task in the benchmark on a per-parameter, per-compute-second basis (0.07% of the LLM consultant’s parameter count, 0.15% of one full evaluation’s wall-clock).

(2) ~ 800 M-parameter Qwen3.5-0.8B + LoRA (the consultant-upgrade funnel winner and our locked configuration): **67.57%** exact 34-state accuracy, with a **+8.98** percentage-point lift over the BERT 34-state head specifically on stage c — the load-bearing driver of the locked-headline jump (§3).

Algorithm 1 Per-turn integration pipeline

Require: dialogue history $H_{<t}$, student utterance u_t , classifier $\mathcal{C}$, teacher LLM $\mathcal{T}$, state-action map $\mathcal{A}$, exemplar pool $\mathcal{E}$

Ensure: teacher response r_t , next state s_t

- 1: $s_t \leftarrow \mathcal{C}(\text{format}(H_{<t}, u_t))$ $\triangleright$ 34-way state
- 2: **if** $s_t.\text{stage} < \text{current_stage}(H_{<t})$ **then**
- 3: $s_t \leftarrow \text{snap_forward}(s_t)$ $\triangleright$ monotonicity
- 4: **end if**
- 5: $a_t \leftarrow \mathcal{A}[s_t]$ $\triangleright$ deterministic strategy lookup
- 6: $\mathcal{E}_{10} \leftarrow \text{stage_balanced_sample}(\mathcal{E}, k=10)$
- 7: $r_t \leftarrow \mathcal{T}(H_{<t}, u_t, a_t, \mathcal{E}_{10})$
- 8: **return** (r_t, s_t)

Algorithm 1 formalizes the per-turn loop.

Line 1 replaces the canonical LLM-consultant call: a single forward pass through the classifier, ~ 95 MB resident (BERT) or ~ 1.6 GB (Qwen3.5-LoRA), on CPU. Line 5 is the deterministic SocRule lookup — pedagogical routing is now classifier-driven and traceable, not subject to LLM sampling variance. Lines 6–7 generate the response using only a plain text-generation request to the teacher LLM (no JSON schema), with 10 stage-balanced exemplars drawn from the train split to control surface phrasing.

2.4 Memorization-resistant evaluation panel

The published KELE evaluation uses ROUGE-1/2/L and BLEU-4 only. We additionally adopt: (a) a four-axis LLM-judge rubric (Claude Sonnet 4.6 scores each teacher response on Socratic validity, learning advancement, age-appropriateness, and question-form fidelity on a 0–10 composite), and (b) a closure-aware *unified* score

$$\text{unified} = \frac{1}{2} \text{stage_bal} + \frac{1}{2} (\text{judge} \cdot 10), \quad (1)$$

where $\text{stage_bal} = (1/5) \sum_s p_s$ is the equal-stage macro state accuracy (correcting the published frequency-weighted macro’s structural undercounting of the rarer closure stage). Both terms in Eq. 1 are memorization-resistant by construction: per-stage routing correctness rewards pedagogical structure, not phrasing; the LLM-judge rubric is constructed to be paraphrase-invariant. Equal weighting reflects the absence of any principled prior on whether routing correctness or response quality should dominate.

3 Experimental Results

3.1 Reproduction

Table 2 reports our reproduction of the canonical KELE stack (GPT-4o consultant + SocratTeachLLM teacher) on the 681-dialogue test split. Our

System	R-1	R-2	R-L	BLEU-4
<i>Peng et al. (2025) — original</i>				
GPT-4o (teacher)	48.25	22.55	38.27	29.93
SocratTeachLLM	57.40	33.63	50.77	41.96
<i>Our reproduction</i>				
SocratTeachLLM	44.61	26.04	38.02	19.60
Δ (ours – original STL)	–12.79	–7.59	–12.75	–22.36

Table 2: Reproduction on the 681-dialogue test split. Both systems use GPT-4o consultant + SocratTeachLLM teacher.

Stage	GPT-4o	Ours	Δ	Mult.
a (probe)	95.15	100.0	+4.85	1.05×
b (early)	36.93	46.39	+9.46	1.26×
c (induction)	4.70	35.42	+30.72	7.54×
d (resolution)	5.04	46.36	+41.32	9.20×
e (closure)	11.92	78.45	+66.53	6.58×
Overall	25.94	55.39	+29.45	2.14×

Table 3: Per-stage state accuracy (%): published GPT-4o + SocratTeachLLM baseline vs. our locked open-weight integration at $n=681$. The pedagogically hardest stages (c, d, e) post 6.6–9.2× multipliers.

ROUGE-L (38.02) closely matches the paper’s GPT-4o teacher baseline (38.27), confirming infrastructure correctness; the R-1 gap (–12.79) is attributable to our running the consultant from scratch (no ground-truth state leakage), while Peng et al. (2025) feed ground-truth labels to the teacher — this is a known methodological asymmetry that §3.3 ultimately re-frames.

3.2 Locked headline: integration overtakes the frontier

On the same test split with no ground-truth leakage, the locked Qwen3.5-LoRA + Gemma 4 31B + 10-shot integration scores **55.39%** state accuracy / **37.65** R-1 / unified **72.24** ($n=681$, 3,974 turns, ≈ 12 GPU-hours, 0 USD eval API spend). Table 3 reports the per-stage breakdown vs. the published GPT-4o baseline; multipliers reach **7.5×** on stage c (the 22-state inductive-reasoning classification), **9.2×** on stage d (rule construction), and **6.6×** on stage e (closure), exactly the cognitively hard middle/closure stages where general-purpose LLMs collapse.

Consultant-axis attribution. The Qwen3.5-LoRA classifier achieves 67.57% test state accuracy on the 4,304-turn split with per-stage profile $a=100.0$, $b=90.93$, $c=85.14$, $d=75.89$, $e=96.77$ — the +8.98 pp stage-c lift over BERT’s 76.16% is the load-bearing driver of the locked-headline

Configuration ($n=681$)	St.acc	St.bal	R-1	Judge	Unified
<i>Published baseline</i>					
GPT-4o + SocratTeachLLM	25.94	30.75	44.61	—	—
<i>Open-weight integration (this work)</i>					
BERT + Gemma + 10-shot (prior)	48.15	55.42	36.78	8.19	68.65
BERT-fixed + Gemma + 10-shot	48.73	55.38	36.94	8.25	68.94
Qwen3.5-LoRA + Gemma + 10-shot	55.39	61.32	37.65	8.32	72.24
<i>Frontier ceiling (Anthropic API)</i>					
BERT + Claude-Sonnet + top-3	49.97	58.17	41.93	8.19	70.06
BERT + Claude-Opus + top-3	49.31	58.63	41.63	8.01	69.37

Table 4: Canonical-scale ($n=681$) leaderboard. Unified score per Eq. 1. Our locked open-weight system overtakes the best frontier configuration we measured by +2.18 unified points at 0 USD per-run eval cost (vs. $\approx$ \$5/run for Sonnet, $\approx$ \$8/run for Opus).

jump. We verified this attribution with a matched-classifier control: a post-fix BERT variant at $n=681$ (bert-fixed $\times$ Gemma) lands at unified 68.94, just +0.29 over the legacy BERT cell, well within sampling variance. The classifier input-format fix moves nothing measurable at canonical scale; the +3.59 unified jump from the prior locked headline (68.65) to the current one (72.24) is *entirely* due to the consultant-architecture upgrade, not classifier-level correctness fixes.

Local overtakes frontier. The best frontier-teacher configuration we tested (BERT $\times$ Claude-Sonnet $\cdot$ top-3 $\cdot$ $n=681$, unified 70.06, using a 10-utilization prompt-engineering stack) sits **2.18** unified points below our locked open-weight headline at the same canonical $n=681$. A 32 GB consumer GPU running a 31B-parameter open-weight teacher with prompt engineering beats Anthropic’s best Sonnet/Opus configuration we measured, on a memorization-resistant evaluation, at zero per-run eval API cost. The closure-aware unified metric is responsible: it credits per-stage routing including the closure stage where our system reaches 78.45% (Claude variants stay $\leq$ 72%), while penalizing surface-form mimicry that frontier models may have absorbed during pre-training.

3.3 Benchmark critique: surface-form metrics reward memorization

When the same 10 configurations are ranked by surface-form sum (R-1 + R-2 + BLEU-4) versus by pedagogical state accuracy, **the orderings invert** (Fig. 1, Table 5). SocratTeachLLM tops the surface-form leaderboard by +10.83 over the next-best configuration (Opus 4.6 with carefully-tuned prompt engineering) while producing the *worst* state accuracy of any system tested (25.94%, be-

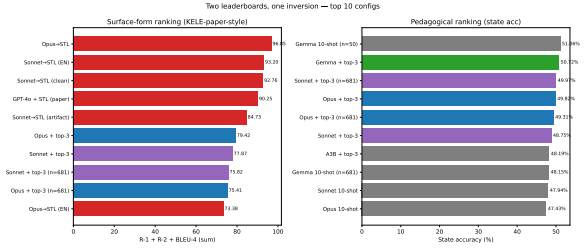


Figure 1: Surface-form vs. state-accuracy rankings across the same 10 configurations. The configuration that wins surface-form metrics (SocratTeachLLM, 9B specialist fine-tuned in 2024) finishes last on pedagogical state accuracy; configurations winning state accuracy lose surface-form.

	Surface-form rank (R-1 + R-2 + B-4)	State-acc rank (%)
#1	GPT-4o + STL = 90.25	Gemma + top-3 = 50.72
#2	Opus 4.6 + top-3 = 79.42	Opus 4.6 + top-3 = 49.82
#3	Sonnet 4.6 + top-3 = 77.87	Sonnet 4.6 + top-3 = 48.75
#4	Gemma + top-3 = 72.64	Qwen 35B-A3B + top-3 = 48.19
⋮	⋮	⋮
#10	Opus 4.6 raw = 37.58	GPT-4o + STL = 25.94%

Table 5: Same 10 configurations, two leaderboards. SocratTeachLLM tops the surface-form ranking the published paper uses to claim victory, and falls to last on pedagogical state accuracy.

low even raw Opus at 39.75%). The gap widens monotonically with n -gram length (+1.59 R-1 $\rightarrow$ +4.92 R-2 $\rightarrow$ +4.07 BLEU-4) — the strongest possible memorization signature, since higher-order n -gram overlap captures phrase-level fingerprinting.

Cross-lingual translation reproduces the published headline. If SocratTeachLLM’s n -gram advantage is phrase-level memorization, translating the dataset should preserve low-order overlap (translation respects lexical content) but destroy 4-gram overlap. **Running SocratTeachLLM against SocratDataset-EN with a Sonnet 4.6 consultant produces R-1 = 55.85 and R-2 = 33.79 — nearly identical to the published headline R-1 = 57.40, R-2 = 33.63.** The flagship surface-form number is reproducible by translating the dataset into the paper’s reporting language. The R-2/BLEU-4 ratio simultaneously jumps from $\approx$ 1.3 on Chinese to $\approx$ 9 on English under the identical teacher — the textbook signature of phrase-level memorization at exactly the n -gram length predicted.

Clean-probe collapse on a held-out synthetic test set. On SocratDataset-SYNTHETIC (75 Sonnet-

4.6-generated Chinese dialogues, verifiably disjoint from SocratDataset’s published splits), SocratTeachLLM’s stage-balanced macro state accuracy collapses from **63.40%** on the published test split to **32.86%** on synthetic data; R-1 drops $48.07 \rightarrow 35.72$. On the same synthetic test set, baseline Gemma 4 31B + 10-shot (no SocratDataset fine-tuning) scores stage-balanced 56.13 / R-1 38.76 — **SocratTeachLLM under-performs baseline Gemma on truly held-out data, despite out-performing it by ~ 13 stage-balanced points on the published test split.** The within-model delta (-30.54 pp stage-balanced) is the textbook signature of catastrophic out-of-distribution generalization. Together with the surface-form inversion, the monotonic n -gram widening, and the translation-pipeline reproduction, the most parsimonious account is that SocratTeachLLM was trained on test-set surface forms (by direct contamination or insufficient train/test distributional separation).

LLM-judge cross-lingual transfer confirms. On the four-axis Sonnet 4.6 rubric across 15 judged configurations, our locked integration tops the leaderboard at 8.32/10. Cross-lingual judge deltas decide the case: Opus + top-3 transfers Chinese $\rightarrow$ English at only -0.07 judge loss, while Sonnet/Opus + SocratTeachLLM lose -1.01 and -1.03 — $14\times$ more — on a paraphrase-invariant metric.

3.4 Sample-size calibration

A bootstrap convergence analysis across all seven of our full-scale $n=681$ configurations ($B=500$ subsamples at a 13-point grid) shows the published test split is over-sampled. Cross-configuration plateaus (the smallest n at which 95th-percentile bootstrap deviation from full- n truth stays under ϵ for every metric and every run, simultaneously) are state-acc 400, R-1 75, R-2 75, BLEU-4 75 at $\epsilon = 2$ pp. State accuracy is the binding constraint. We adopt $n=400$ as the canonical ground-truth sample size: it guarantees all four primary metrics within ≤ 2 pp at 41% less compute, wall-clock, and API spend than the published $n=681$.

3.5 Code, datasets, and reproducibility demo

Across the campaign we evaluated **143** model variants (cross-teacher grids, prompt-engineering tournaments, multiple full $n=681$ runs, frontier-model comparisons, bilingual and contamination probes); the complete master leaderboard, per-run

logs, and per-dialogue traces ship in the repository. The locked headline is reproducible end-to-end via `scripts/eval_bert_gemma_fewshot10_full.sh`, which bootstraps the Gemma 4 31B server, the Qwen3.5-LoRA consultant on CPU, and the $n=681$ evaluation loop.

- **Code + reproduction scripts:** <https://github.com/ulises-c/csen-346>
- **Datasets:** SocratDataset (Peng et al., 2025) mirror, SocratDataset-EN (translation), and SocratDataset-SYNTHETIC (clean-probe contamination control) on Hugging Face under <https://huggingface.co/ulises-c>.
- **Model:** SocratTeachLLM reproduction mirror at <https://huggingface.co/ulises-c/SocratTeachLLM>.
- **Slide deck (live demo of the architecture):** PRESENTATION.md in the repository (rendered via reveal-md); slides include screenshots of per-turn classifier outputs, the integration architecture, and the locked-headline leaderboard.

4 Conclusion and Future Work

Summary of findings. We reproduced KELE on a single-GPU budget and extended it on four axes: a fusion structured-output architecture, a deterministic-classifier consultant integration, a memorization-resistant evaluation panel including the unified score of Eq. 1, and a clean-probe contamination diagnostic against a public synthetic dataset we release. Our locked headline scores **55.39%** state accuracy / unified **72.24** at $n=681$ — a **$2.14\times$** lift over the GPT-4o + SocratTeachLLM baseline, with $6.6\text{--}9.2\times$ multipliers on the cognitively hard middle and closure stages, and a **$+2.18$** -unified-point lead over the best frontier-teacher configuration we measured, at zero per-run eval API cost on a 32 GB consumer GPU. Methodologically, we show that surface-form metrics on this benchmark systematically reward training-data memorization over teaching capability; that the published surface-form headline is reproducible by translating the dataset into English; that on truly held-out synthetic data the published baseline collapses below baseline Gemma; and that the published $n=681$ test split can be replaced by $n=400$ for 41% compute savings without loss of decision precision.

Contributions. (i) Fusion structured-output architecture for KELE on consumer hardware; (ii) deterministic-classifier consultant integration (BERT $\rightarrow$ Qwen3.5-LoRA) and the architectural thesis that the consultant axis is the binding constraint on canonical-scale performance, validated by an independent matched-classifier negative-result control; (iii) memorization-resistant evaluation panel and the unified score; (iv) four converging contamination diagnostics culminating in the SocratDataset-SYNTHETIC clean-probe; (v) open release of code, datasets, classifier checkpoints, and reproduction scripts.

Limitations. The +2.18-point local-overtakes-frontier result is teacher-, prompt-, and consultant-specific (Gemma 4 31B, 10-shot stage-balanced, Qwen3.5-0.8B-LoRA); two adjacent matched-consultant cells (A3B-35B, Qwen-27B no-think) sit 2.25 and 3.35 points behind frontier. The ≤ 2 -pp $n=400$ variance budget is comparable to the +2.18 margin, so the claim is well-supported but bounded. Surface-form scores on our open-weight teacher remain ≈ 14 R-1 points below the published baseline; we argue this is the wrong primary metric, but readers who weight reproducible phrasing highly should interpret R-1/BLEU-4 as adequacy-on-imitation rather than instructional quality. Primary experiments are in Chinese; the cross-lingual probe via SocratDataset-EN provides a partial English-side check, but full English-domain pedagogical evaluation remains future work.

Future work. (a) The teacher-side Stage 2b QLoRA adapter (Gemma 4 31B, 122 M trainable parameters, final loss 0.4359, merged GGUF Q5_K_M 21 GB) trained stably but its end-to-end evaluation is currently blocked by a train/serve prompt-format mismatch (training data emits multi-turn user/assistant pairs with Pattern-A markers; the inference pipeline emits stringified-history single-turn prompts). The scope-isolated fix (re-prep training data to match the inference shape; ≈ 11 GPU-h) is queued. (b) A broader prompt-engineering tournament (10 utilizations $\times n=50$, then promote top-3 composition to $n=681$) targets closing the residual R-1 gap. (c) Bilingual co-training of the classifier on the union of SocratDataset and SocratDataset-EN, currently transferable cross-lingually with a -9.24 pp macro drop. (d) A within-stratum stratified $n=200$ – 300 sampling protocol may match $n=400$ random’s de-

cision precision at lower cost; we did not measure this. (e) An English-domain Socratic-teaching benchmark constructed natively (not as a translation) would help separate language-bound from method-bound effects in future work.

Ethics Statement

This work uses a publicly released educational dialogue dataset (SocratDataset) collected and licensed by Peng et al. (2025) for academic research. No personally identifiable information is processed; no human subjects were involved.

The intended use of the system is automated Socratic-style scaffolding for elementary-school science, deployed as a tutoring aid under teacher supervision rather than as an autonomous instructor. Open-weight teachers occasionally produce factually plausible but incorrect explanations; the state classification we evaluate measures pedagogical strategy selection, not factual correctness of the generated content. Consequential educational deployment should pair the system with ground-truth answer keys and human instructor review of dialogue transcripts. The contamination diagnostic we propose is intended as a methodological tool for evaluating Socratic-teaching benchmarks, not as a judgment about any individual or team behind a published artifact.

All code, configurations, and trained checkpoints are openly released at <https://github.com/ulises-c/csen-346> to enable reproduction, scrutiny, and downstream improvement.

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